

Mechanical Properties of fluids

HYDRO-STATICS

- Fluid - Any thing that can flow e.g. Gases and liquids.

Ideal fluid - (i) incompressible (Fixed density)
(ii) Non-viscous
(iii) Irrotational

Pressure

Thrust acting normally per unit area around the point

$$\left[P = \frac{F_1}{A} \right]$$

Scalar Quantity

SI unit = Nm^{-2} = Pascal, Dimension: $[ML^{-1}T^{-2}]$

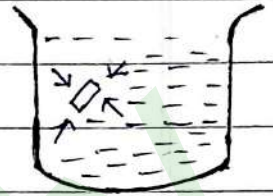
Unit conversion:-

$$1 \text{ atm} = 1.01 \times 10^5 \text{ Pa} \approx 1 \text{ bar} = 760 \text{ mm of hg} = 760 \text{ torr} = 10.33 \text{ m of water}$$

$$1 \text{ torr} = 133 \text{ Pa}.$$

Pressure in fluids

- when an object is submerged in a fluid at rest, the fluid exerts a force on its surface which is always perpendicular to the surface.



Density

- Density is defined as mass per unit volume
Denoted by ρ (rho)

$$\left[\rho = \frac{M}{V} \right]$$

- Positive scalar quantity.

SI unit = kg m^{-3} , cgs = g/cc

- unit conversion of volume -

$$1 \text{ L} = 10^{-3} \text{ m}^3 = 10^3 \text{ cm}^3 = 1 \text{ dm}^3$$

Fluid	ρ
	C kg m^{-3}
Water	1×10^3
Sea water	1.03×10^3
Mercury	13.6×10^3
Ethyle alcohol	0.806×10^3
Whole blood	1.06×10^3
Air	1.29
Oxygen	1.43
Hydrogen	9×10^{-2}
Inter stellar Space	$\approx 10^{-20}$

- Density of a mixture of two or more liquids

[case I] Suppose two liquids of densities ρ_1 and ρ_2 having masses m_1 and m_2 mixed together.

$$\left[\rho_{\text{mix}} = \frac{\text{Total mass}}{\text{Total volume}} = \frac{m_1 + m_2}{\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2}} \right]$$

Case II IF two liquids of density ρ_1 and ρ_2 having volumes V_1 and V_2 are mixed.

$$\left[\rho_{\text{mix}} = \frac{\rho_1 V_1 + \rho_2 V_2}{V_1 + V_2} \right]$$

Relative density / specific gravity

- Ratio of the density of substance to the density of water at 4°C .
- No unit
- water have highest density at 4°C

$$R.D = \frac{\rho_{\text{substance}}}{\rho_{\text{reference (water)}}}$$

- For example, relative density of mercury
 $= \frac{13.6 \times 10^3}{10^3} = 13.6.$

Variation of fluid Pressure with depth

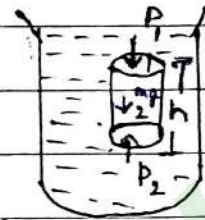
$$F_1 = P_1 A$$

$$F_2 = P_2 A$$

$$W = mg = \text{volume} \times \text{density} \times g$$

$$= \text{Area} \times \text{height} \times \text{density} \times g$$

$$(W = Ahpg)$$



$$P = \frac{Ahpg}{A} = hpg$$

As the fluid is in equilibrium

Net downward force = Net upward force

$$F_1 + W = F_2$$

$$F_2 - F_1 = W$$

$$P_2 A - P_1 A = W$$

$$A(P_2 - P_1) = Ahpg$$

$$P_2 - P_1 = hpg$$

$$[P = P_0 + hpg]$$

at. pressure $\rightarrow$ vertical height

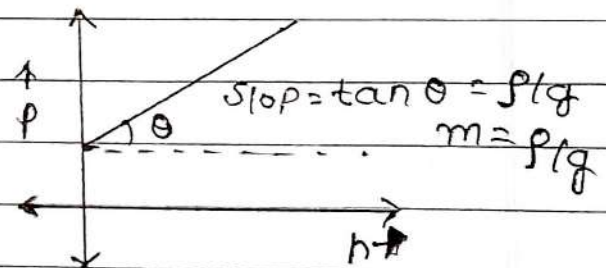
Graph of p and h

$$p = h$$

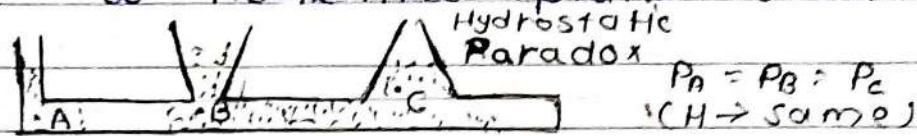
$$p = P_0 + hpg$$

$$p = hpg + P_0$$

$$[y = mx + c]$$



→ For equilibrium, pressure at two points in a horizontal plane is same



→ Pressure at two points which are at a depth separation of h ,

$$P - P_0 = \rho gh$$

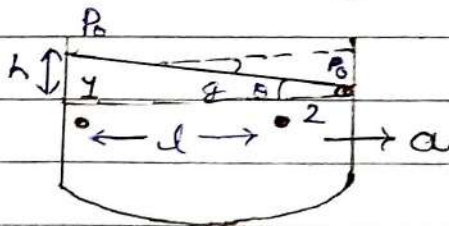
Gauge Pressure and Absolute Pressure

Absolute pressure $P = P_0 + \rho gh$

$\underbrace{P - P_0}_{\Delta P} = \rho gh$

→ P_{Gauge} is measured with manometer.

- Excess pressure (ΔP) at depth h is called a gauge pressure at that point.
- Gauge pressure = Absolute pressure - Atmospheric pressure.
- Pressure when Fluid containing container is accelerating

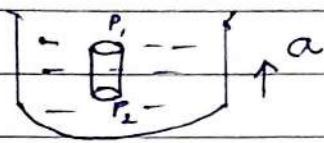


Pressure at two points in a horizontal plane when container is in constant acceleration,

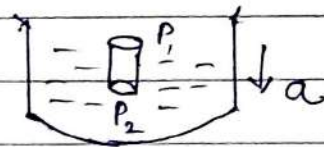
$$[P - P_0 = \rho la]$$

$$\left[\tan \theta = \frac{a}{g} \Rightarrow \frac{h}{d} = \frac{a}{g} \right]$$

- Vertically accelerating fluid



$$[P = P_0 + h\rho(g+a)]$$

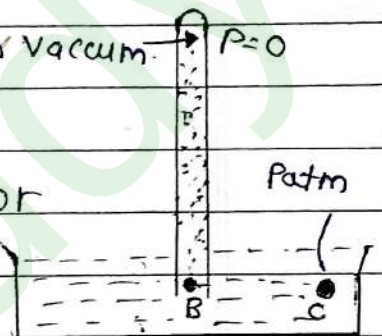


$$[P = P_0 + h\rho(g-a)]$$

Barometer

Torricellian Vacuum $P=0$

- Torricelli devised a method for measuring atmospheric pressure



- A long glass tube closed at one end and filled with mercury is inverted into a trough of mercury.

This device is known as 'mercury barometer'

$$P_B = P_C = P_{atm}$$

$$P_B = 0 + h\rho g$$

$$[P_{atm} = h\rho g]$$

- ★ Height of mercury at sea level is about 76 cm.

$$P_{atm} = h\rho g = \frac{76}{100} \times 13.6 \times 10^3 \times 10 \approx 1.01 \times 10^5 \text{ N/m}^2 \text{ or Pa}$$

$$76 \text{ cm of Hg} = 1 \text{ atm.}$$

$$\approx 1 \text{ atm.}$$

$$[760 \text{ mm of Hg} = 1 \text{ atm}]$$

$$[760 \text{ torr} = 1 \text{ atm}]$$

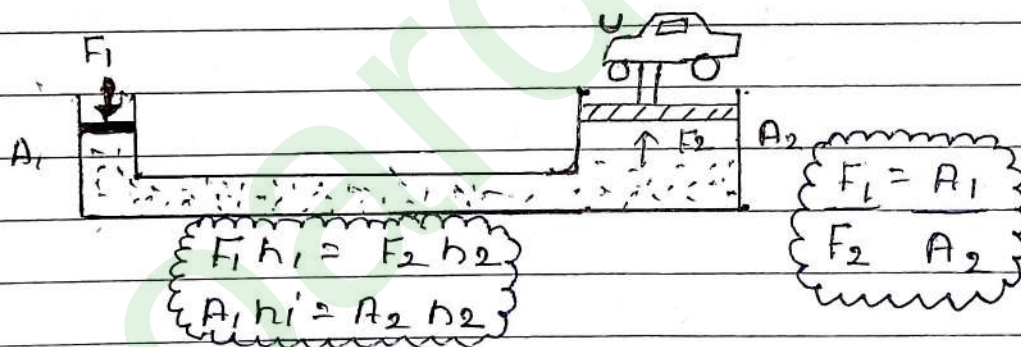
A pressure equal of 1mm is called a torr (after Torricelli)
(1 torr = 133 Pa)

Pascal's law

- Pressure exerted at any point on an enclosed liquid is transmitted equally in all direction.
- If pressure in an enclosed fluid is changed at a particular point, the change is transmitted to every point without being diminished.

Application of pascals law :-

Hydraulic lifts, hydraulic jacks, hydraulic press, hydraulic brakes, etc



Buoyancy

The upward Force acting on a body immersed in a fluid, is called upthrust or buoyant force and phenomenon is called buoyancy.

Archimedes' Principle

- When a body is partially or wholly immersed in a fluid, it is experiences an upward thrust equal to the weight of the fluid displaced by it.

$$\text{Upthrust (U)} = V_f \rho_f g$$

Density of Fluid.

Vol. of fluid displaced
 = Vol. of body immersed
 $U = mfg$ ($m = \rho V$)
 $U = \text{weight of fluid displaced.}$



Apparent weight of body.

- When a body is wholly immersed in a fluid, then its effective weight decreases.
- Apparent weight = Actual weight - Buoyant Force.

$$w_{app} = w_{air} - U$$

$$w_{app} = w_{air} \left[1 - \frac{\rho_{fluid}}{\rho_{solid}} \right] = V \rho g \left[1 - \frac{\rho_{fluid}}{\rho_{solid}} \right]$$

$$\rho_{solid} = \frac{w_{air}}{w_{air} - w_{app}} \times \rho_{fluid}$$

Law of Floatation

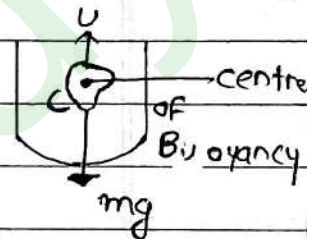
$$V_{submerged} = \frac{\rho_{body}}{\rho_{liquid}} V$$

$$V_{out} = \left[1 - \frac{\rho_{body}}{\rho_{liquid}} \right] V$$

Case I :- Density of body $>$ Density of liquid, Body will sink.

Case II - $\rho_{\text{body}} = \rho_{\text{liquid}}$, Body will float fully immersed (equilibrium)

Case - III :- $\rho_{\text{body}} < \rho_{\text{liquid}}$, Body will float partially immersed.



* A body will float in a liquid if the weight of the liquid displaced by the part of the body is equal to or greater than the weight of the body.

$$F_{\text{net}} = 0$$

$$U = mg$$

$$V_{\text{fluid}} \rho_f g = \rho_{\text{solid}} V_s g \Rightarrow (V_f \rho_f = \rho_s V_s) =$$

$$(h_f \rho_f = h_s \rho_s)$$

↳ not for sphere.

* for two or more liquids

$$\boxed{\rho_1 V_1 + \rho_2 V_2 = \rho_s V_s}$$

or

$$\boxed{\rho_1 h_1 + \rho_2 h_2 = \rho_s h_s}$$

* If two immiscible liquids of mass m_1 and m_2 and density ρ_1 and ρ_2 are mixed

$$\rho_{mix} = \frac{M_{mix}}{V_{mix}} = \frac{m_1 + m_2}{\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2}}$$

If liquids are of same mass, $m_1 = m_2 = m$,

$$\rho_{mix} = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2}$$

* If a body is weighed in air (w_A), in water (w_w) and in a liquid (w_L) then

$$\text{Specific gravity of liquids} = \frac{\text{loss of weight in liquid}}{\text{loss of weight in water}} = \frac{w_A - w_L}{w_A - w_w}$$

HYDRO-DYNAMICS

- When a fluid moves in such a way that there are relative motions among the fluid particles, the fluid is said to be following

Types of fluid flow

1. Steady flow / Streamlined flow

→ Types of flow in which the fluid velocity at a point do not change with time.

- Path Followed in a steady Flow is called as Streamline.
- At different point, velocities of fluid particles can be different.
- Velocity of fluid particles at any point of Streamline is along tangent to the Curve at that point.



Properties of Streamline Flow

- Streamlines do not intersect each other.
- Greater is the crowding in streamline → Greater is the velocity.
- Steady flow is achieved at low speeds. Beyond a limiting value, called critical speed this flow loses steadiness and becomes turbulent.

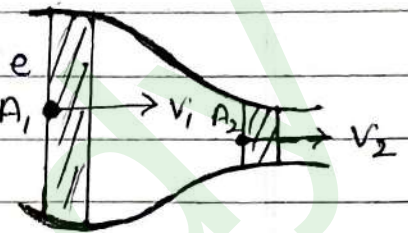
2. Laminar and turbulent Flow.

- Flow in which fluid particles move along well-defined streamlines which are straight & parallel ⇒ Laminar flow
- Flow in which particles move in a zig-zag way ⇒ Turbulent flow.

Equation of continuity

→ mathematical expression of the law of conservation of mass in fluid dynamics.

→ fluid should be incompressible
(Density fixed)



$$A_1 v_1 = A_2 v_2 = \text{Flow Rate}$$

Mass of fluid entering per second = mass of fluid leaving per second

$$\rho \times V_{\text{inside}} = \rho \times V_{\text{outside}}$$

Per second Per second

$$[A_1 v_1 = A_2 v_2]$$

$$t = 1 \text{ s}$$

$$d = v \times t = v$$

$$Av = \text{constant}$$

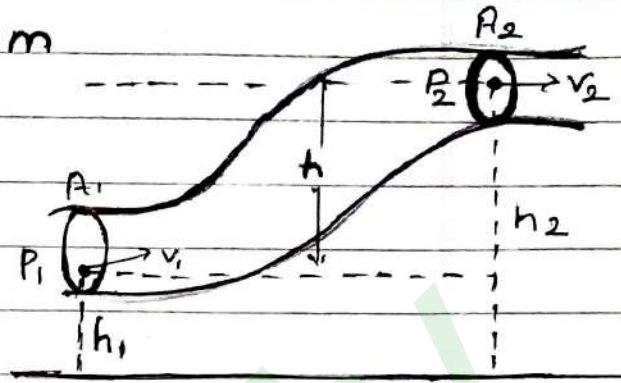
$$(v \propto \frac{1}{A})$$

- Volume flux (Rate of flow)
Volume that flows per second

$$\left[Q = \frac{dv}{dt} = Av \right]$$

➤ Bernoulli's Theorem

- Applicable to ideal fluid + incompressible, non-viscous, steady, irrotational



- KE per unit volume

$$\frac{K.E}{\text{Volume}} = \frac{1}{2} \frac{mv^2}{V} = \frac{1}{2} \rho v^2$$

- Potential energy per unit Volume

$$\frac{P.E}{\text{Volume}} = \frac{m}{V} gh = \rho gh$$

- Pressure energy per unit volume

$$\frac{\text{Pressure}}{\text{Volume}} = P$$

- According to Bernoulli's theorem in case of steady flow of incompressible and non-viscous fluid through a tube of non-uniform cross-section then the sum of all pressure, P.E and K.E per unit volume is same at every point in tube.

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho gh_2$$

$$\left[P + \frac{1}{2} \rho v^2 + \rho gh = \text{constant} \right]$$

→ It expresses conservation of mechanical energy in case of moving fluids.

Different heads

Divide the above eqⁿ by ρg , we get

$$\frac{P}{\rho g} + \underbrace{\frac{v^2}{2g}}_{\text{velocity head}} + \underbrace{h}_{\text{Potential Head / Gravitational Head}} = \text{constant}$$

Pressure head

→ All heads have dimensions of length.

► For fluid at Rest ($v_1 = v_2 = 0$)

$$P_1 + \rho g h_1 = P_2 + \rho g h_2$$

$$P_1 - P_2 = \rho g h_2 - \rho g h_1 = \rho g (h_2 - h_1)$$

$$\boxed{P_1 - P_2 = \rho g h}$$

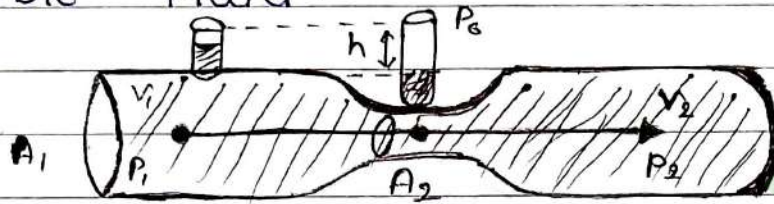
► In horizontal fluid ($h_1 = h_2$)

$$\boxed{P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2} \quad A_1 v_1 = A_2 v_2 \Rightarrow \left[v \propto \frac{1}{A} \right]$$

Applications of Bernoulli's Theorem

• Venturimeter / Flowmeter

→ Device to measure the flow speed of incompressible fluid.



Same Horizontal level

$$(A_1 V_1 = A_2 V_2)$$

$$P_1 + \frac{1}{2} \rho V_1^2 = P_2 + \frac{1}{2} \rho V_2^2$$

$$P_1 = P_0 + \rho g h$$

$$P_1 - P_2 = \frac{1}{2} \rho (V_2^2 - V_1^2) \quad \text{--- (i)}$$

$$P_2 = P_0 + \rho g h_2$$

$$P_1 - P_2 = \rho g (h_1 - h_2) = \rho g h.$$

$$P_1 - P_2 = \frac{1}{2} \rho g h \quad \text{--- (ii)}$$

Equating (i) & (ii) we get

$$(V_2^2 = V_1^2 + 2gh)$$

3. Rate of flow = $A_1 V_1 = A_2 V_2$

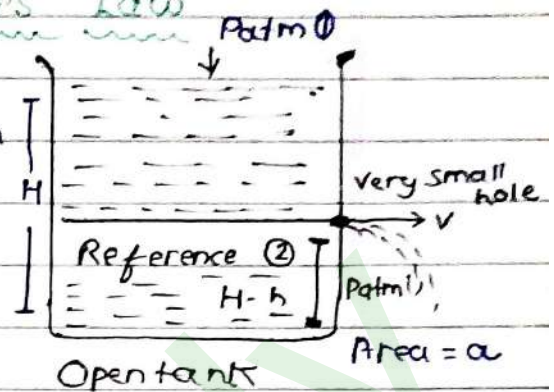
$$\left\{ Q = A_1 A_2 \sqrt{\frac{2gh}{A_1^2 - A_2^2}} \right\}$$

Speed of Efflux : Torricelli's Law

Applying Bernoulli's Theorem

$$P_{atm} + \frac{1}{2} \rho (0)^2 + \rho gh =$$

$$P_{atm} + \frac{1}{2} \rho v^2 + \rho g(0)$$



$$\rho gh = \frac{1}{2} \rho v^2$$

$$v^2 = 2gh \quad \text{! - valid for open tank}$$

$$v = \sqrt{2gh} \quad \text{! - Same as free fall}$$

Rate of Flow (Q)

$$(Q = A \times v = a v = a \sqrt{2gh})$$

Range

$$F = P_{av}$$

y-direction

x-direction

$$S = v \times t$$

$$R = \sqrt{2gh} \times \sqrt{\frac{2g(H-h)}{g}}$$

$$R = 2\sqrt{h(H-h)}$$

$$\text{Range} = T \cdot v$$

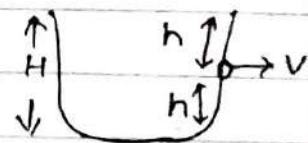
So the range

of water is $(R = 2\sqrt{h(H-h)})$

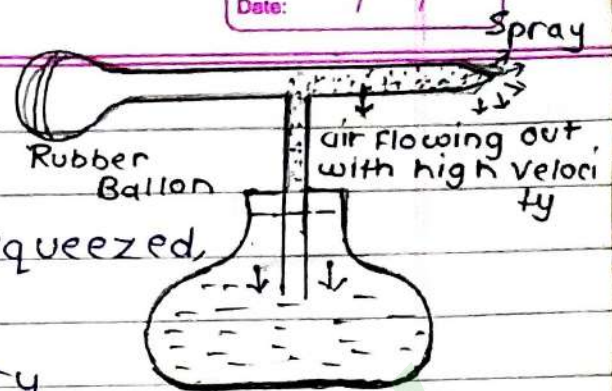
Maximum Range

$$R_{max} \rightarrow h = \frac{H}{2}$$

$$R = 2\sqrt{\frac{H}{2} \left(\frac{H-H}{2} \right)} = H$$



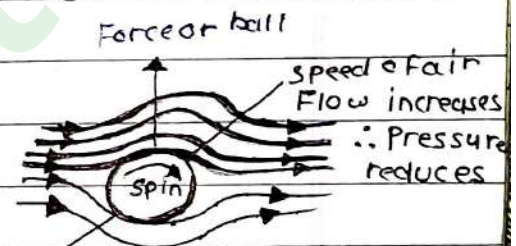
Atomizer or Sprayer



When rubber ball is squeezed, air rushes out through horizontal tube with very high velocity and thus the pressure reduces. The liquids in vessel rises up and mixes with air in form of small droplets which gets ejected in form of fine spray.

e.g. Paint guns, perfume or deodorant spray etc.

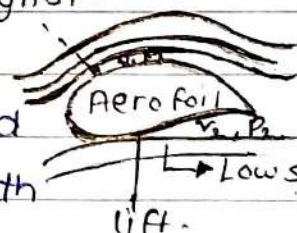
• Magnus effect



A/Q to Bernoulli's principle the pressure on the upper side becomes lower than that on lower side. This pressure difference exerts a force on the ball to which it moves along curved path.
 Speed of air flow decreases $\therefore$ Pressure increased
 Speed of air flow increases $\therefore$ Pressure reduces

• Aerofoil

Pressure above reduced whereas that underneath it get increased.



$$F = (P_2 - P_1) A$$

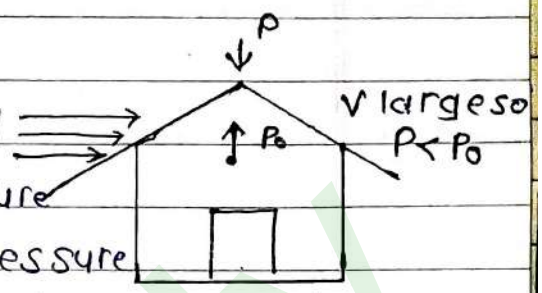
$$= \frac{1}{2} \rho (v_1^2 - v_2^2) A$$

Low speed, more pressure
 lift.

Thus, a resultant upward thrust is generated normal to the flow and it is this force which provides most of the upward lift for an aeroplane.

e.g. Aircraft wings, turbine blades and propellers.

• Blowing-off of Tin Roof Tops in wind storm

when wind flows with high velocity above a tin roof, wind  It causes lowering of pressure above roof. Due to this pressure difference, the roof is lifted up and is blown away during storms.

Pull-in or Attract force by fast moving trains

→ Same reasons

• Blood flow and Heart attack

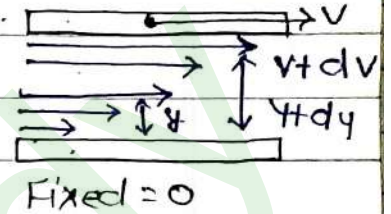
In advance heart condition, artery gets constricted due to accumulation of plaque. To remove it speed of blood increases in this region → Pressure decreases (Heart Attack) ← Artery may collapse

Viscosity (fluid friction)

- No ideal fluid.

- Viscosity is internal friction in a fluid motion.
- It occurs due to relative motion between layer of liquid.
- The internal friction force in fluid.
⇒ viscous force.

- Tangential to layers / Shearing.
- Viscous Force depends on relative motion between layers (velocity gradient), area of contact b/w two layer and type of fluid.



- Velocity Gradient = $\frac{\Delta v}{\Delta y} = \frac{dv}{dy}$
- For any layer of liquid, its upper layer pulls it forward while lower layer pulls it backward. This results in the Force between the layers. This type of flow is known as Laminar

$F_v \propto$ contact area of layer.

$F_v \propto$ Velocity gradient b/w two layer.

$$F_v \propto A \frac{dv}{dy} \Rightarrow F_v = -\eta A \frac{dv}{dy}$$

Speed $\rightarrow$
height $\rightarrow$

• Coefficient of viscosity

- $\eta \rightarrow$ Property of a fluid $\eta = \frac{F/A}{v/l}$

$\eta \uparrow \rightarrow$ extent of viscous force

- SI unit : $N \cdot s \cdot m^{-2} = Pa \cdot s = \text{poiseuille (PI)} = \text{deca poise}$

- CGS unit : $\text{dyn} \cdot s / \text{cm}^2 = \text{poise}$.

- Dimension = $[ML^{-1}T^{-1}]$ (1 deca poise = 10 poise)

* The viscosity of liquids decreases with temperature, while it increases in case of gases.

* Velocity for viscous flow in pipe is maximum at centre.

Stoke's Law (For spherical body)

If a small sphere of radius r is moving with a velocity (v) through a homogeneous stationary medium of viscosity η then the viscous force acting on sphere is,

$$(F_v = 6\pi\eta r v)$$

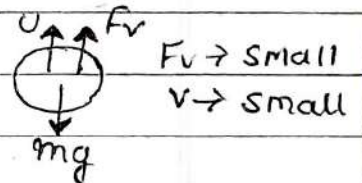
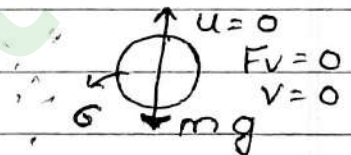


Force will always be opposite to velocity. $\text{air drag} \rightarrow \text{viscous force}.$

• Terminal Velocity

At equilibrium, $F_{\text{net}} = 0$

$a = 0$, $v = \text{constant} = \text{Terminal velocity}.$



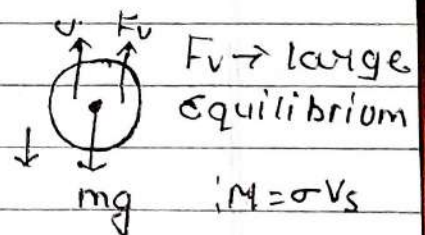
$$U + F_v = F_g$$

$$V_f \rho_f g + 6\pi\eta r v_T = \sigma V_s g$$

$$\frac{4}{3}\pi r^3 \rho g + 6\pi\eta r v_T = \sigma \frac{4}{3}\pi r^3 g$$

$$36\pi\eta r v_T = \frac{4}{3}\pi r^3 g (\sigma - \rho)$$

$$3\eta v_T = \frac{2}{3} r^2 g (\sigma - \rho)$$

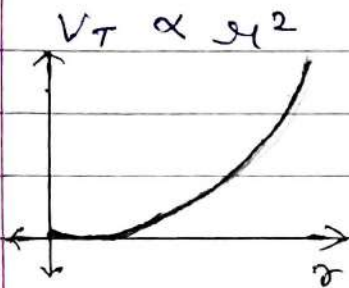


$$v_T = \frac{2}{9} \frac{r^2 g (\sigma - \rho)}{\eta}$$

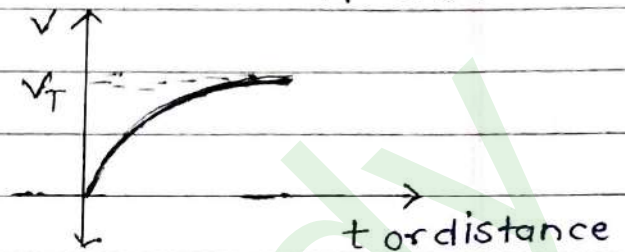
$\sigma = \text{density of sphere}$
 $\rho = \text{density of fluid}.$

Graph

V_T V/s x



V V/s t for Freely Falling Sphere.



- If density of sphere is less than that of fluid ($\sigma < \rho$)
 $V_T = -V_e$ e.g. Air bubble in water rises up.

→ Body moves upward.

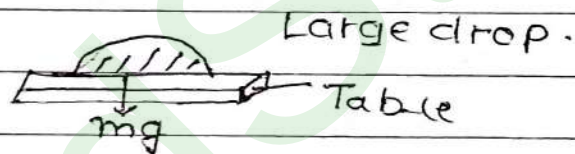
Some Application of Terminal Velocity

- Actual Velocity of rain drops is very small in comparison to velocity acquired by a body falling freely from height of cloud.
- Decent of a parachute with moderate velocity.
- Determination of electronic charge in millikan's oil drop experiment.

Surface Tension

- Surface Tension is the property of liquid Surface. (elastic behaviour)
- The property of liquid surface due to which it tries to attain minimum surface area is called surface tension

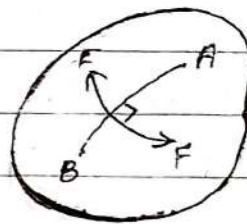
E.g - A small drop of liquid is spherical in shape, because For a give volume, sphere, has min. surface Area.



- Surface Tension is numerically equal to the Force acting per unit length of a hypothetical line drawn on a liquid Surface normal to it in either direction

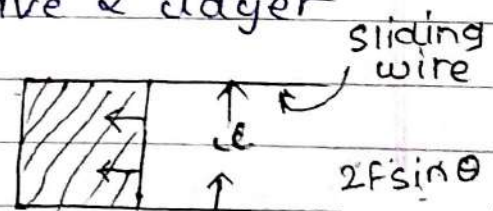
$$T = \frac{F}{2} \left(\frac{N}{m} \right)$$

$$F = T \ell$$



- Liquid Film $\rightarrow$ Always have 2 layer

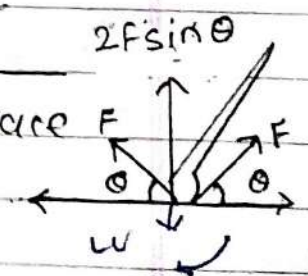
$$F = T \times (2\ell)$$



- Floating of Needle on water surface

$$2F \sin \theta = w$$

$$\Rightarrow w = 2(T\ell) \sin \theta$$



$$F = \int T dl, F = T \ell$$

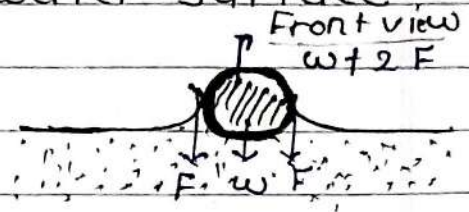
$\theta \rightarrow$ angle of contact.

- Lifting of Needle above water Surface

$$F_{\text{excess}} = 2Tl$$

$$= w + 2F$$

$$F_{\text{min}} = w + 2(2Tl)$$



Some important point :-

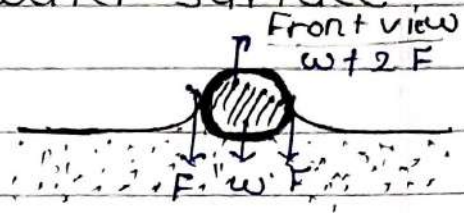
1. Soap decreases the surface tension of water, which helps soap water to reach its pores of clothes and takes out dirt.
2. Surface Tension of a liquid decreases with rise in temperature and becomes zero at critical temperature, that is why hot soup tastes better than cold one.
3. Surface Tension is a scalar quantity. (However direction of Force on line on which it is calculated can be known)
4. It depends upon nature of liquid and its intermolecular Forces.
5. Depends on impurities added to a liquid. (decreases T) (insoluble impurity)
6. Surface Tension of water is maximum at 4°C .

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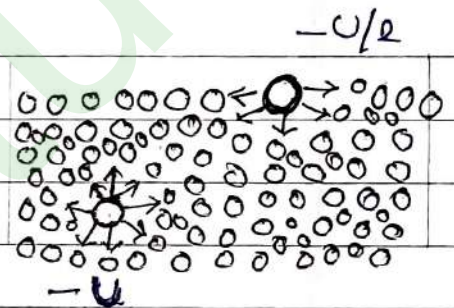
- Required excess force For a circular thick ring having internal and external radii r_1 & r_2 , $F = F_1 + F_2 = T(2\pi r_1) + T(2\pi r_2)$
 $[F_{\text{excess}} = 2\pi T (r_1 + r_2)]$

- If $r_1 = r_2 = r$ $[F = 4\pi r T]$

- For a circular disc, $[F = T \times 2\pi r]$

- Surface Energy

- molecule on a liquid surface have some extra energy in comparison to molecules in the interior.



- Addition energy of surface - Surface energy
 A liquid, thus, tends to have least surface to have minimum energy and most stability.

Intermolecular Force of attraction = P.E = Negative.

- Surface energy is nearly half of heat of Vapourisation.

Relation b/w Surface Energy and Surface Tension (U)

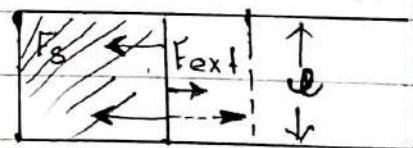
$$U \propto \text{Surface area}, \quad \Delta U = T \Delta A \quad \left| \quad T = \frac{\Delta U}{\Delta A} = \frac{\gamma}{M^2} \right.$$

$$W_{\text{ext}} = F_{\text{ext}} \cdot x$$

$$= F_s \cdot x$$

$$= T(2l)x$$

$$= T \underbrace{2(lx)}_{\text{Total change in Area}}$$



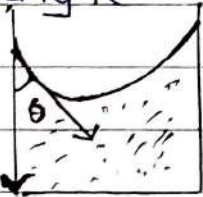
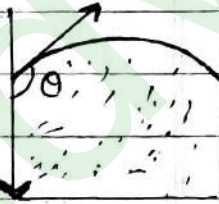
$F_s = T(2l)$ very slowly Sided x .

Total change in Area

$$\Delta U = T \Delta A$$

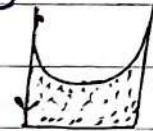
Angle of contact (θ)

The angle between tangent to the liquid surface at the point of contact and tangent to solid surface, measured inside the liquid is termed as contact angle.

$\theta < 90^\circ$ → concave meniscus → Liquid Rises upwards → wettability increases (Expand)		Impure water $\theta > 90^\circ$ → convex meniscus → Liquid Fall down toward → Wettability decreases.		mercury
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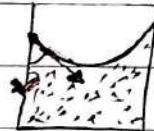
→ For pure water and perfectly clean glass, the angle of contact is 0°

- Complete wet [$\theta = 0^\circ$]



→ For ordinary water and glass, it lies b/w 8° and 18°

- High wettability ($\theta < 90^\circ$)



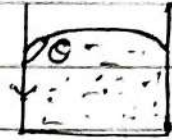
→ For pure water and pure silver, the angle of contact is 90°

Low wettability

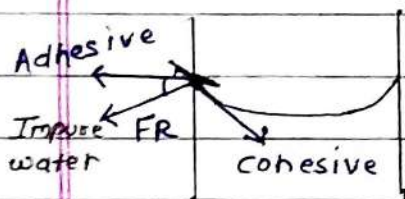


No meniscus

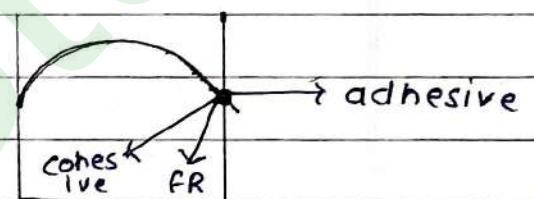
→ For mercury and perfectly clean glass, the angle of contact is 135°
 ($\theta > 90^\circ$) Bad wettability



- Cohesive Force: Force of attraction b/w molecules of same kind.
 [More cohesive force = Strong T]
- Adhesive Force: Force of attraction b/w molecules of different kind.



Adhesive > cohesive



Cohesive > Adhesive

→ Liquid Surface is always Normal to Resultant Force.

Surface Tension $\uparrow = 0 \downarrow$

Drops and Bubbles

⇒ Number of surfaces in liquid film and bubble.

1. Liquid Film / ^{layers} 2 surface (Always)

2. Bubble / drop. only soap have 2 surface, other wise 1 surface

• Excess Pressure in a liquid Drop

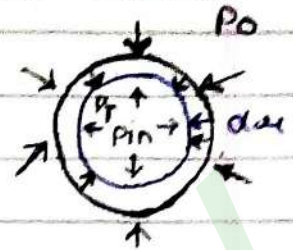
Liquid drop experiences compressive stress due to:

- P_0 (outside Pressure)
- due to surface tension (P_T)

At equilibrium

$$P_{in} = P_0 + P_T$$

$$\boxed{P_{in} = P_0 + \frac{2T}{R}} \Rightarrow \Delta P = \frac{2T}{R} \quad (P_{in} > P_0)$$



$$\rightarrow P_{excess} = P_{in} - P_0$$

Radius of drop changed from r to $(r + dr)$, then

$$\text{work done} = F \cdot dr$$

$$= (P_{ex} A) dr$$

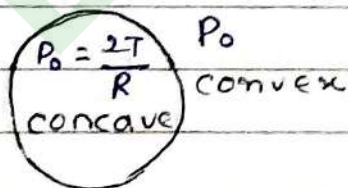
$$= P_{ex} 4\pi r^2 dr \quad dr^2 \text{ is negligible}$$

$$\text{change in surface area} = 4\pi (r + dr)^2 - 4\pi r^2 = 8\pi r dr$$

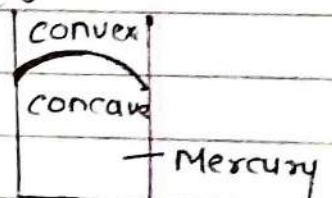
$$T = \frac{W}{\Delta A} = \frac{P_{ex} 4\pi r^2 dr}{8\pi r dr} = \frac{P_{ex} r}{2}$$

$$\Rightarrow \Delta P = \frac{2T}{r}$$

• Excess pressure on concave side:



$$P_{internal} > P_{ext}$$

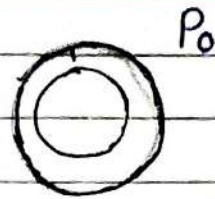


$$P_{in} - P_0 = \frac{2T}{R}$$

$$P_{concave} - P_{convex} = \frac{2T}{R} \quad (P_{concave} > P_{convex})$$

- For a soap bubble $\rightarrow$ 2 Surface

$$P_{in} - P_0 = \frac{4T}{R}$$



- Excess pressure inside a cavity or air bubble in liquid -

$$P_{excess} = P_{in} - P_{atm}$$

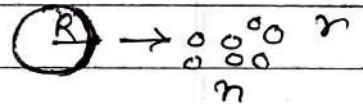
$$= \rho gh - \frac{2T}{R} \quad (\rho = \text{density of liquid})$$

$$\begin{array}{c} P_{atm} \\ \uparrow h \\ P_{in} \end{array}$$

$$P_{inside} = P_{atm} + \rho gh + \frac{2T}{R}$$

- Splitting of water drop in small droplets

Volume remain conserved



$$\textcircled{1} \quad \frac{4}{3} \pi R^3 = n \left[\frac{4}{3} \pi r^3 \right]$$

$$\textcircled{2} \quad r = \frac{R}{n^{1/3}}$$

$$\textcircled{3} \quad W = E = 4\pi R^3 \left[\frac{1}{r} - \frac{1}{R} \right] T$$

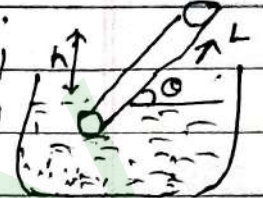
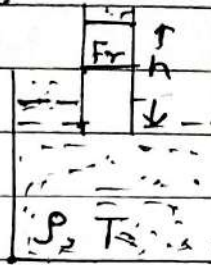
• Capillarity

The phenomenon of rise or fall of a liquid in the capillary is called capillarity.

> Height of capillary Rise

$$h = \frac{2T \cos \theta}{\rho g r}$$

$$p = \frac{2T \cos \theta}{r}$$



h = Vertical height of rise or Fall.

r = Radius of tube.

$h = \frac{2T \cos \theta}{\rho g r}$

$$\theta < 90^\circ$$

$$\cos \theta = +ve$$

$$h = +ve, \text{ Rise}$$

Concave meniscus

$$\theta = 90^\circ$$

$$\cos \theta = 0$$

$$h = 0$$

No meniscus

$$\theta > 90^\circ$$

$$\cos \theta = -ve$$

$$h = -ve, \text{ Fall}$$

Convex meniscus

Radius of tube (r) and Radius of meniscus (R)

$$\cos \theta = \frac{r}{R}$$

$$(R \cos \theta = r)$$



$$h = \frac{2T \cos \theta}{\rho g r} = \frac{2T \cos \theta}{R \cos \theta \rho g}$$

$$h = \frac{2T}{R \rho g}$$